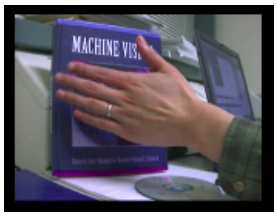
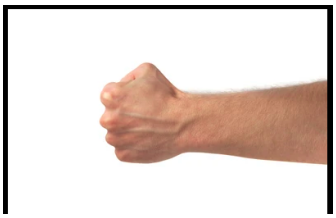
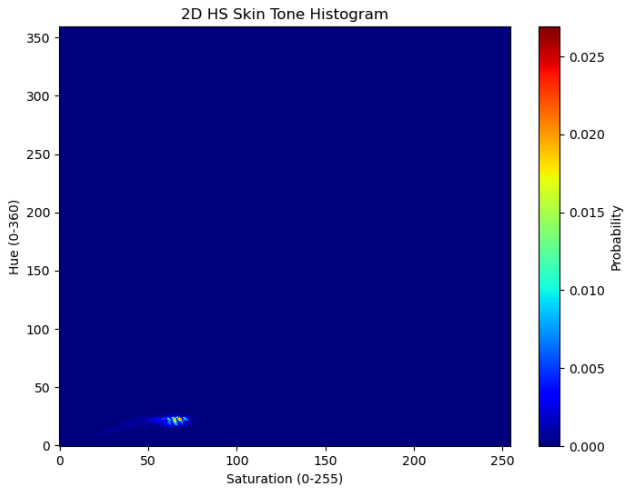
Nolan Knight
MSAI495
MP#3
April 15, 2026

In this problem, we applied color segmentation using 2D histograms to isolate skin pigment in images. The following sample images were used for the training and testing:

As shown in the image, each of the hand's skin pigmentation is slightly different, or is within a different lighting. This means we have to isolate skin pigmentation under different conditions and ranges. The goal of the color segmentation was to gather training data by isolating known skin pixels from training data, and then construct a 2D histogram tone detector to determine skin pigment pixels. In order to gather training data, I isolated a square in the image, and converted the cropped section of the image into an array, where I then added each pixel to a master list of all training pixels. To construct the 2D tone detector, I chose to use HSI coordinates, as I felt that they would best represent the distribution for the skin pigments. This is because we want to focus on the Hue and Saturation relationship of our skin pigments without worrying about color intensity (different lighting conditions).

Using this data, I was able to build a 2Dimensional relationship between Hue and Saturation for the extracted training data. The following 2D Histogram was created:



As shown, there is a stronger relationship between a saturation value between 20-70 and a hue value of approximately 10-30. These values represent the combination of the hue and saturation that best represents the skin pigments. After constructing the 2D histogram tone detector, where I used the threshold value of 0, or if there is a positive correlation between H and S, then the image pixel is segmented. Using this technique I generated the following segmentation images from the testing set:

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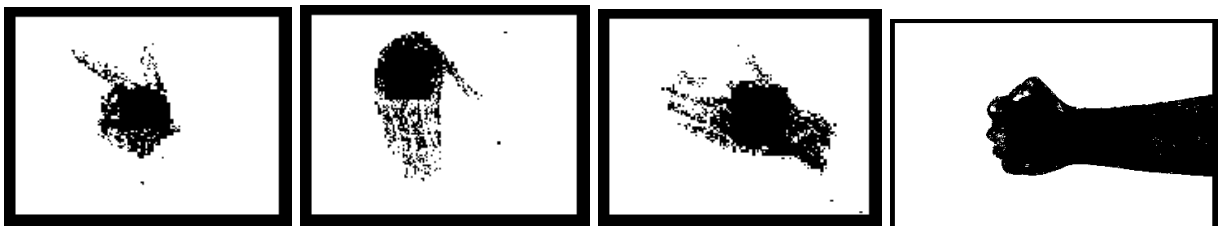
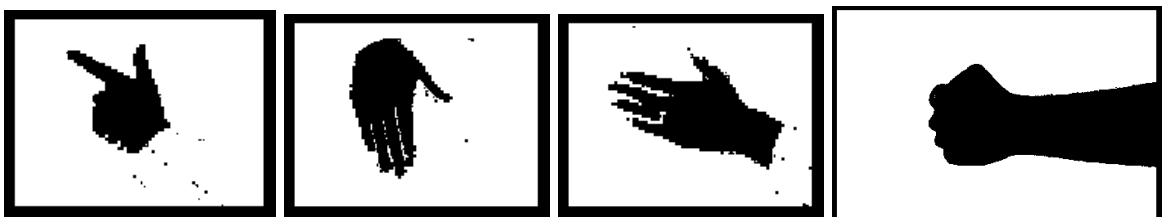
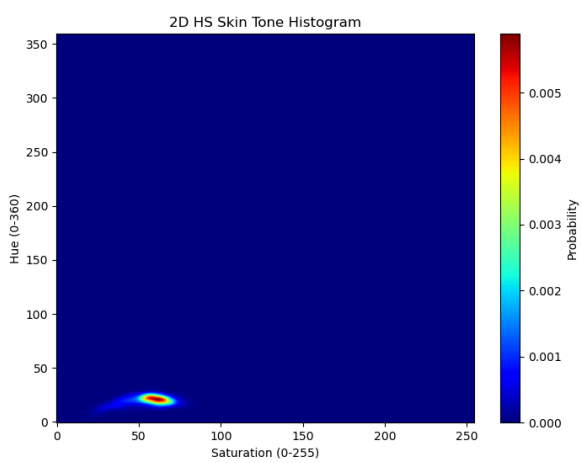


With this output, I found that there was an issue with isolating all of the relationships between hue and saturation to isolate the majority of the skin pixels. I assumed this was either from not an appropriate amount of training data, leading to some of the relationships being ignored in the 2D Histogram. Therefore, I applied a gaussian smoothing filter to spread the relationship of the 2D HS Histogram to pick up probabilities of some of the neighboring relationships. This resulted in the following image segmentation for each of the test images:



As a result, I determined that applying the smoothing filter to the histogram tone detector was beneficial as it picked up on most of the missing skin pigments missed by the original trained detector. This helped pick up on many tones that may have been missed or shadowed within the images. After the applied gaussian filter to the HS-Histogram, the HS-histogram:



As shown it better spreads the probability of the combination of H and S being a skin tone, giving a larger spread of relationships with a probability above 0.